OpenClaw Temperature and Humidity Dashboard Tutorial

> Before starting, please complete the following from the `Pre-use Preparation` directory: [01-Peripheral Hardware Configuration], [02-OpenClaw API-KEY Configuration], [03-OpenClaw Model Switching], [05-AI Voice Interaction Configuration], and [06-Three Dialogue Configuration Tests]. If you are currently only using TUI / WhatsApp, you can skip `05`. This article assumes the basic environment is already prepared and only covers the Temperature and Humidity Dashboard project itself.

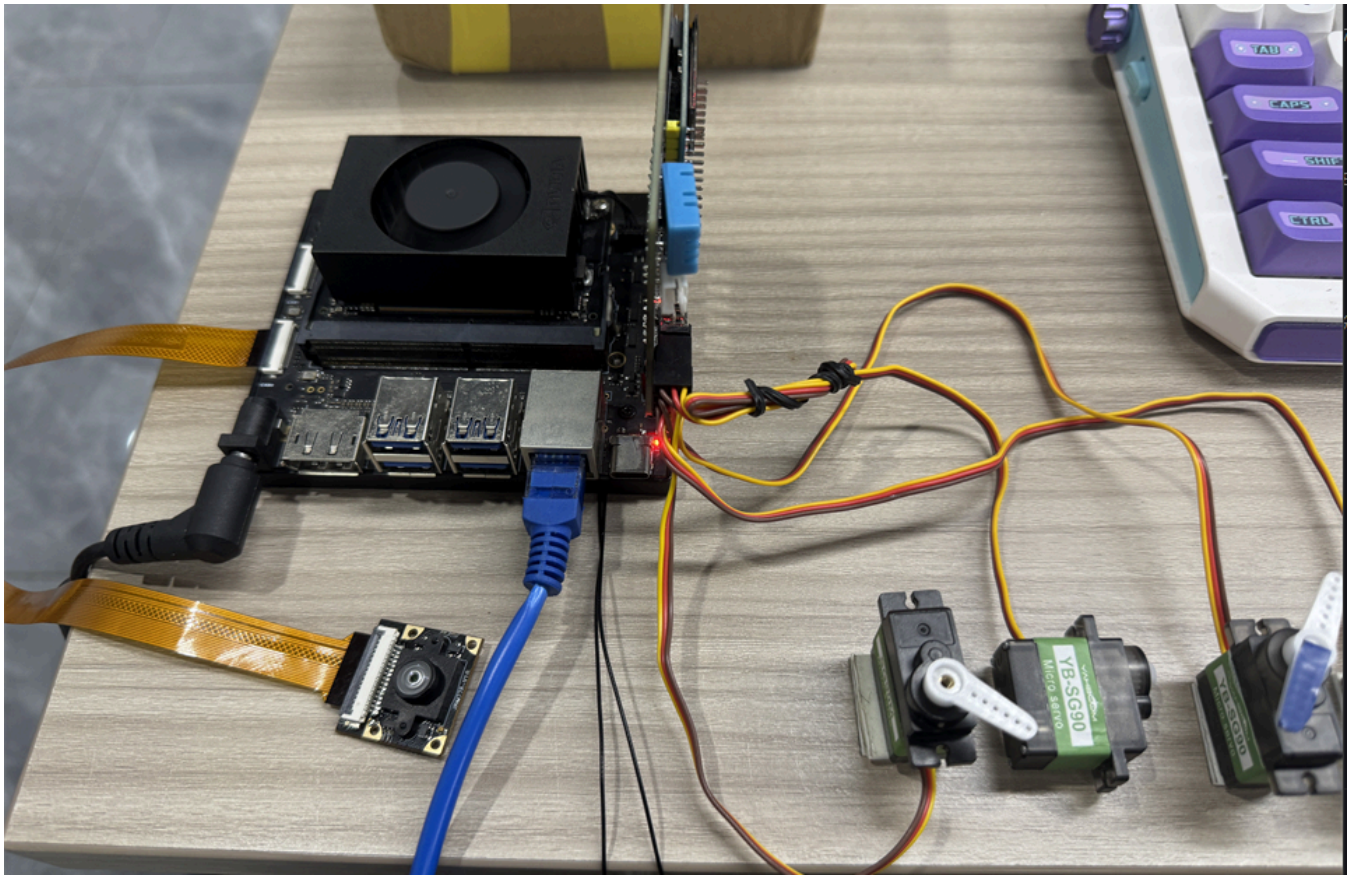
1. Tutorial Objectives

This article demonstrates OpenClaw's multi-peripheral coordination capabilities in a desktop dashboard scenario through the combination of "sensor + display + indicator". After completing this tutorial, readers will be able to:

- Understand how to combine DHT, OLED, RGB, and servo into a temperature and humidity dashboard
- Interact with OpenClaw through three methods: WhatsApp, TUI, and voice code
- Complete four project scenarios: real-time readings, OLED indicator panel, RGB status light, and servo pointer mapping

2. Hardware Preparation

- OpenClaw main control board
- DHT temperature and humidity sensor
- OLED display
- RGB light strip / RGB LED module
- `[Optional] AI voice module + speaker`
- Wiring diagram



3. Software Preparation

Code entry points used in this tutorial

- Temperature and humidity reading command: `/home/jetson/hardware_hal_demo/dht`
- OLED control command: `/home/jetson/hardware_hal_demo/oled`
- RGB control command: `/home/jetson/hardware_hal_demo/rgb`
- Servo control command: `/home/jetson/hardware_hal_demo/cli/servo`
- Voice entry: `/home/jetson/voice_to_openclaw/src/main.py`

This project is very similar to the AI Weather Station, but with a different focus:

- The AI Weather Station emphasizes "broadcasting" and "interpretation"
- The Temperature and Humidity Dashboard emphasizes "display" and "status mapping"

4. Testing Project Basic Capabilities

Before starting this project, please complete a basic `openclaw tui` dialogue self-check; if you haven't done so yet, see [06-Three Dialogue Configuration Tests] first. After passing, proceed to the Temperature and Humidity Dashboard specific test.

It is recommended to first enter OpenClaw TUI:

```
openclaw tui
```



Then test the following types of commands in sequence:

- Read current temperature and humidity
- Display current temperature and humidity on OLED
- If the environment is normal, turn on green light
- Turn the servo to 90 degrees

5. Three Solutions for Communicating with OpenClaw

Solution	Suitable Scenarios	Advantages	Recommended Use
WhatsApp	Remote status viewing, classroom demonstrations	Convenient for remotely reading current panel data	Demonstrate "remote monitoring panel"
TUI	Local debugging, mapping verification	Easiest for troubleshooting data flow between sensors and actuators	Run through the visualization pipeline first
Voice code	Voice broadcast, desktop assistant	Closer to daily usage methods	Use as the complete demonstration solution

If the optional AI voice module + speaker is selected, Feishu dialogue, TUI, and WhatsApp dialogue need to play back OpenClaw's replies via TTS. You can run the following program in the terminal. The AI voice module dialogue itself does not need to run it again.

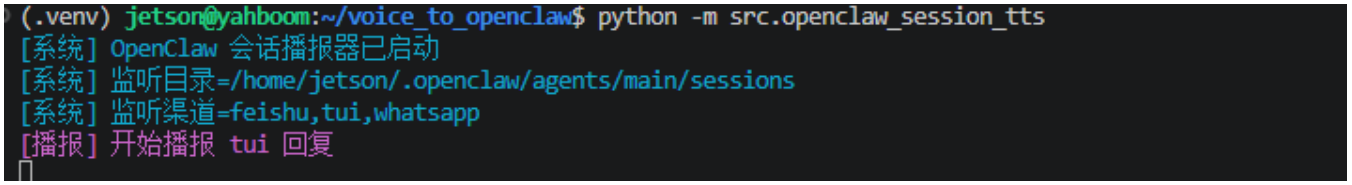
- Modify the TTS broadcast parameter

```
/home/jetson/voice_to_openclaw/.env
```

```
ENABLE_TTS=true  # This parameter controls whether the replies are broadcast. After modification, save it and run the following program in the terminal
```

- Enter the following in the terminal

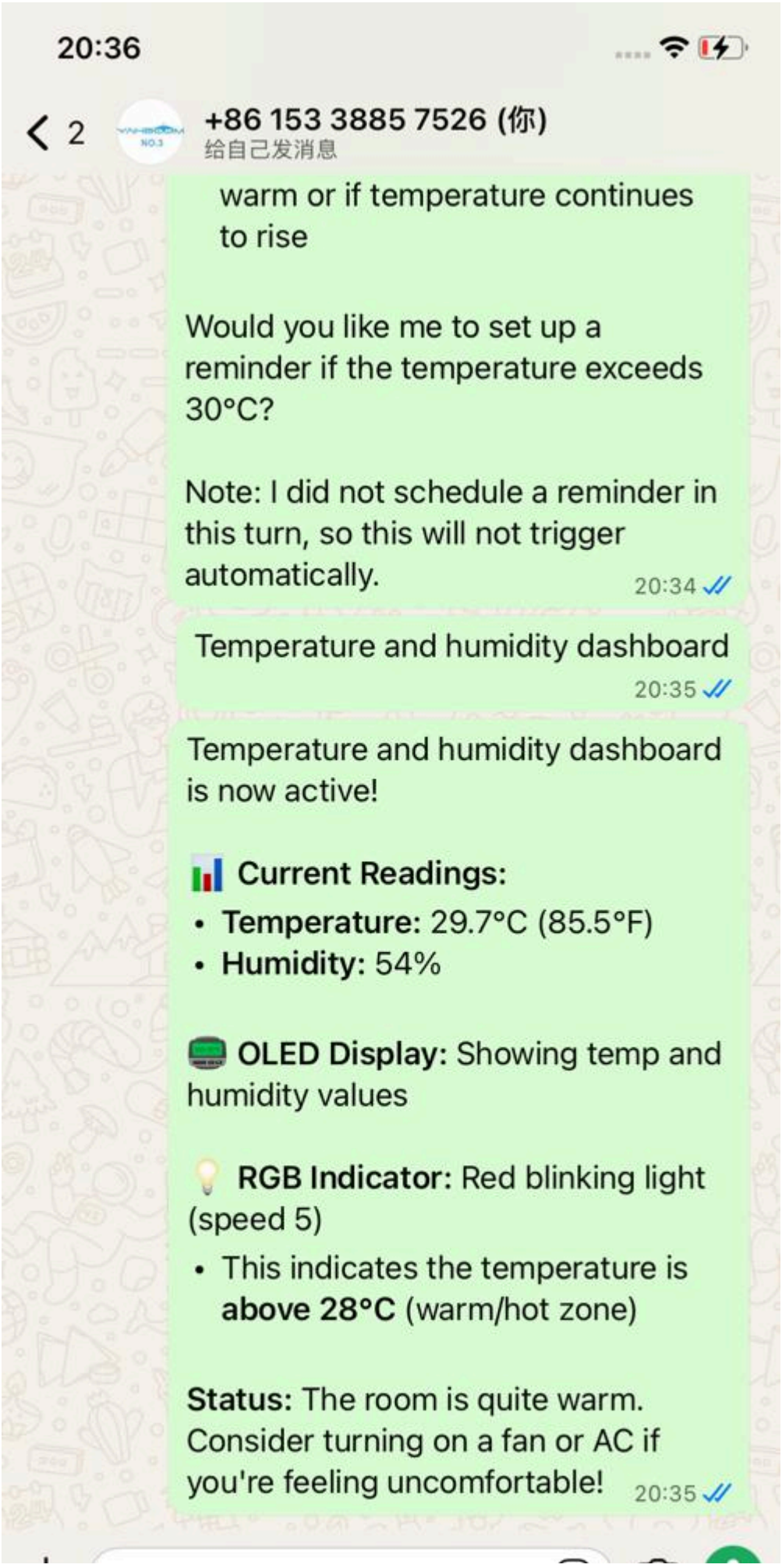
```
cd ~/voice_to_openclaw
source .venv/bin/activate
python -m src.openclaw_session_tts
```



5.1 WhatsApp Solution

For WhatsApp integration and bot creation steps, please refer directly to [06-Three Dialogue Configuration Tests]. After creation, you can remotely view dashboard status in WhatsApp, for example:

- Temperature and humidity dashboard


20:36
2
+86 153 3885 7526 (你)
给自己发消息
warm or if temperature continues to rise
Would you like me to set up a reminder if the temperature exceeds 30°C?
Note: I did not schedule a reminder in this turn, so this will not trigger automatically.
20:34
Temperature and humidity dashboard
20:35
Temperature and humidity dashboard is now active!
Current Readings:
Temperature: 29.7°C (85.5°F)
Humidity: 54%
OLED Display: Showing temp and humidity values
RGB Indicator: Red blinking light (speed 5)
This indicates the temperature is above 28°C (warm/hot zone)
Status: The room is quite warm. Consider turning on a fan or AC if you're feeling uncomfortable!
20:35



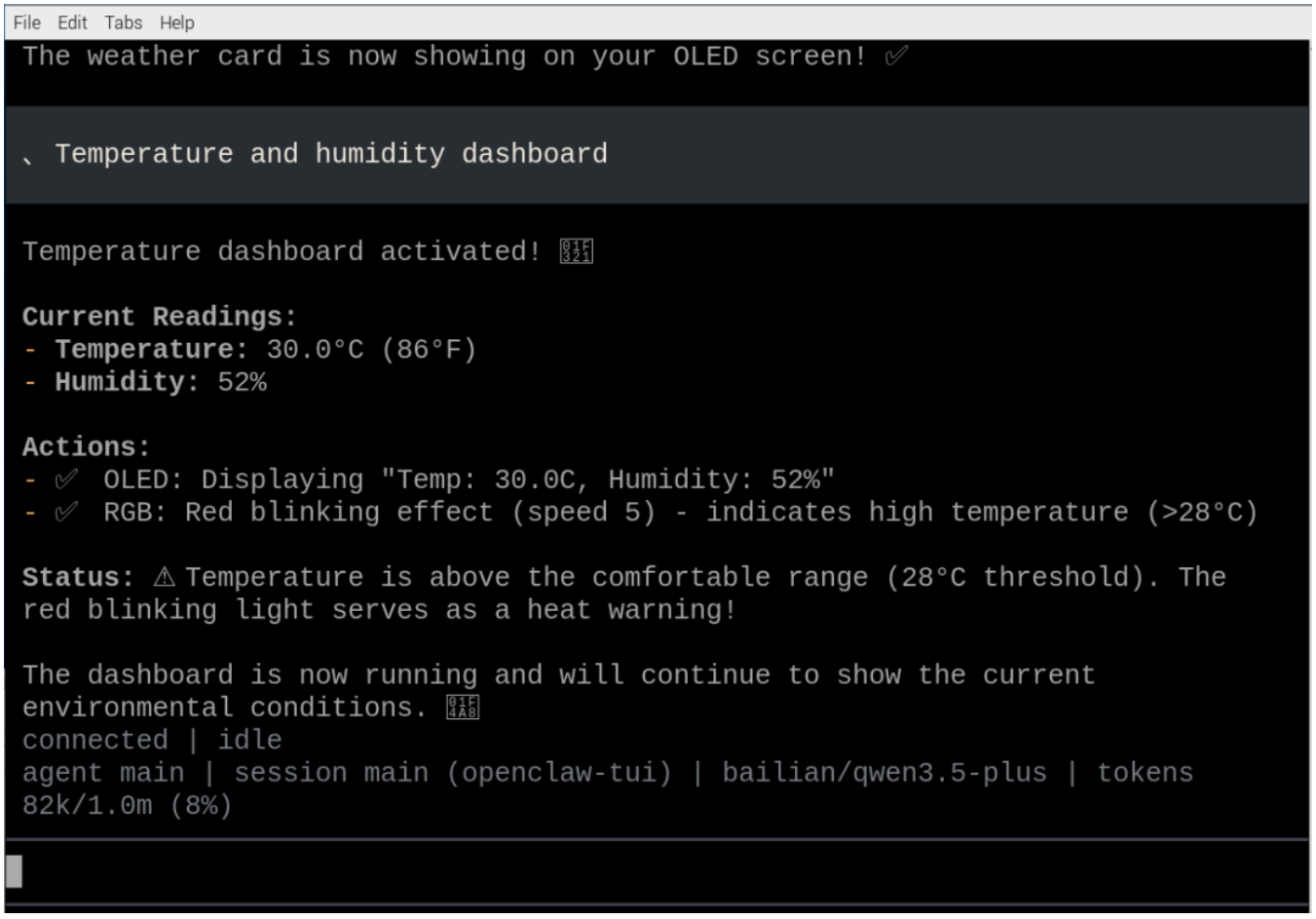
5.2 TUI Solution

If you have completed the TUI self-check in article 0, directly enter terminal text dialogue:

```
openclaw tui
```

Then enter project-related commands, for example:

- Temperature and humidity dashboard

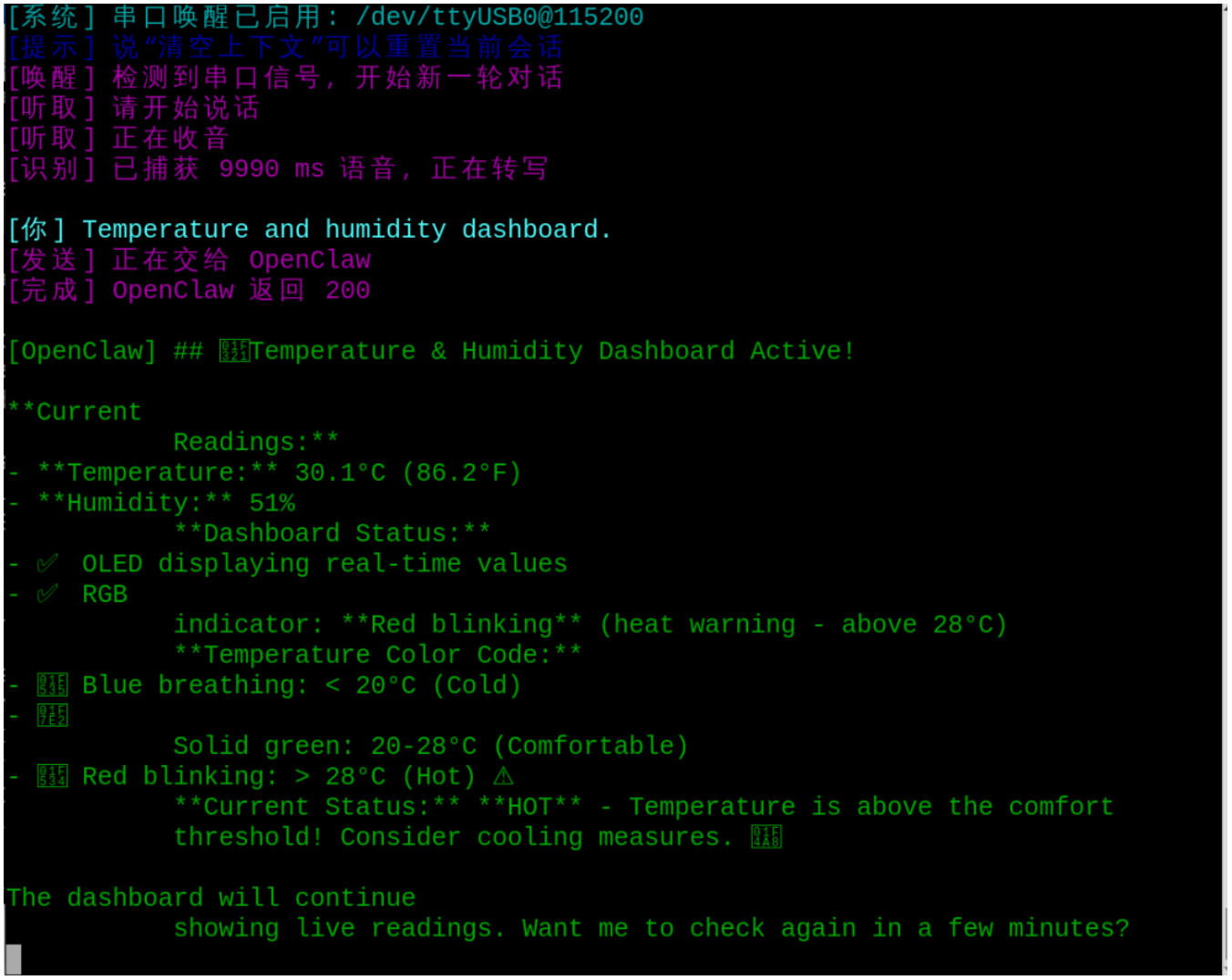


5.3 Voice Code Solution

```
Requires optional AI voice interaction module + speaker
```

For voice module installation, environment configuration, and basic wake-up procedure, please refer to [05-AI Voice Interaction Configuration]. After wake-up, you can directly speak control commands, for example:

- Temperature and humidity dashboard



6. Project Implementation

Note: The following demonstrations are all conducted using `openclaw tui` dialogue. You can also choose WhatsApp or voice methods to complete them.

6.1 Scenario 1: Real-time Readings

Experiment Objective

Read the current temperature and humidity to complete the basic data input for the dashboard.

Effect Description

OpenClaw will first retrieve the temperature and humidity readings, which serve as the basis for subsequent OLED, RGB, and servo mapping.

Example Commands

- Read current temperature and humidity
- What's the current temperature and humidity
- Help me refresh the current readings

6.2 Scenario 2: OLED Indicator Panel

Experiment Objective

Display the current temperature and humidity results as concise OLED dashboard content.

Effect Description

This is suitable for using `metrics` or `dht_oled` methods to combine `Temp` and `Hum` into a compact screen area, creating a small data panel.

Example Commands

- Display Temp and Hum on OLED
- Create a simple temperature and humidity panel
- Write the current temperature and humidity to the screen

6.3 Scenario 3: RGB Status Light

Experiment Objective

Use RGB colors to express whether the current environment is normal, hot, cold, or dry.

Effect Description

This step can make the dashboard more intuitive. For example, green for normal, yellow for hot, red for abnormal, blue for cold — you can see the status at a glance.

Example Commands

- If the environment is normal, turn on green light
- If the temperature is high, turn on yellow light
- If the humidity is abnormal, turn on red light

6.4 Scenario 4: Servo Pointer Mapping

Experiment Objective

Map temperature or humidity to servo angle, making the servo point to a position like a gauge pointer.

Effect Description

For example, map the temperature range 0 - 50 to 0 - 180 degrees. The higher the current temperature, the farther right the servo pointer moves. This effect is very suitable for classroom demonstrations, because "turning numbers into physical actions" is very intuitive.

Example Commands

- Use the servo to represent current temperature like a pointer
- Map the current temperature to servo angle
- Make the pointer point to the current status